THEORY

Technology affects real people

Use the five-issue comparison to identify each impact, then explain its effects on people or society.

Issue	What it means (simple)	Example of digital technology	Impact on society
Ethical	What is right or wrong when using technology	AI facial recognition	Could reduce crime but may unfairly target groups
Legal	Laws that control how technology is used	Copyright on YouTube	People can be fined for illegal downloads
Cultural	How technology changes behaviour and lifestyle	Social media	Changes how people communicate
Environmental	How technology affects the planet	Data centres	Uses lots of electricity
Privacy	How personal data is collected and used	Smartphones tracking location	People lose control of personal information

FROM EXAMPLE TO EXPLANATION

Ethical: facial-recognition bias may unfairly affect groups. Legal: copying content may break copyright rules. Cultural: social media may change communication. Environmental: data centres consume energy and resources. Privacy: location tracking can expose personal information.

EXAM HABIT

For longer questions, connect the technology to an issue, explain an effect, and reach a reasoned result. Avoid answers made only of a list of issue names.

THEORY · TIER

Use the TIER writing framework

TIER stands for Technology, Issues, Effects and Result. Use these four stages to structure an extended answer.

Students lose marks in longer questions because they **list** instead of **discuss** always use **TIER** to avoid **tears**
T–I–E–R (Technology → Issues → Effects → Result)

Think in TIER. Write in paragraphs. Never list. If the question says **discuss**, aim for **6 marks** by using more than one (> 1) issue and explaining the > 1 effects.

TIER STEPS

TIER Steps	What it means	Helpful sentence starter	Example (Social Media)
T – Technology	The digital technology you are talking about	This technology is...	Social media
I – Issues	The problems the technology causes (more than one)	This causes problems because... and...	People share too much personal information and behaviour online changes
E – Effects	What happens to people or society because of the issues	This can lead to... and...	People lose privacy and mental health or behaviour can be affected
R – Result	What happens next because of the effects	As a result...	Privacy settings, rules and online safety awareness are used

HOW TIER EARNS MARKS

T: identify the technology. I: name the issues requested by the question. E: explain consequences for real people or organisations. R: give a reasoned result or judgement. In an extended answer, linked explanations and balance earn marks; the acronym alone does not.

THEORY · WORKED MODEL

How TIER becomes an answer

Study this model of a social-media answer, with the thinking structure beside a written student answer.

Question (6 marks)	TIER (Thinking Structure)	Student exam answer
Discuss the ethical, cultural and privacy issues caused by the use of social media.	**T** – Social media lets people talk and share things online. **I** – This causes problems because: personal data is collected (privacy issue) AND online behaviour changes (cultural issue). **E** – This can lead to: people losing privacy AND people sharing too much information and spending less time talking face to face, which can affect mental health. **R** – As a result, privacy settings, rules and online safety awareness are used.	Social media causes privacy and cultural issues because personal data is collected and online behaviour changes. This can lead to people losing privacy and sharing too much information, as well as spending less time talking face to face, which can affect mental health. As a result, privacy settings and online safety rules are used.

WHAT TO NOTICE

The worked table contains an issue (privacy and changed social behaviour), an effect (loss of control over data and changed communication), then a result (better choices and safeguards). The final answer makes these links in prose.

IMPROVE THE EXAM ARGUMENT

Add balance when asked to discuss: online communication can connect distant families, but data collection and pressure to post also have drawbacks. Explain both rather than repeating only disadvantages.

THEORY · MORE ANALOGIES

Worked examples using TIER

The guide applies TIER to everyday technology and legal/ethical issues, not just social media.

Discuss the legal and ethical issues caused by online shopping websites.	**T** – Online shopping lets people buy items on the internet. **I** – This causes problems because:• customer rights must be protected by law (legal issue)AND• personal data is collected and used (ethical issue). **E** – This can lead to:• customers being tricked or treated unfairlyAND• personal data being misused. **R** – As a result, consumer laws and secure payment systems are used.	Online shopping causes legal and ethical issues because customer rights must be protected and personal data is collected. This can lead to customers being tricked and personal data being misused. As a result, consumer protection laws and secure payment systems are used.
Discuss the ethical and privacy issues caused by artificial intelligence in everyday life.	**T** – Artificial intelligence is used to make choices and analyse data. **I** – This causes problems because:• AI decisions may be unfair (ethical issue)AND• large amounts of personal data are collected (privacy issue). **E** – This can lead to:• people being treated unfairlyAND• people losing control of their personal information. **R** – As a result, rules and controls are introduced to manage AI use.	Artificial intelligence causes ethical and privacy issues because decisions may be unfair and lots of personal data is collected. This can lead to people being treated unfairly and losing control of their information. As a result, rules and controls are introduced to manage how AI is used.

TECHNICAL ACCURACY CHECK

For online shopping, the issue is not simply that collecting personal data is illegal. Businesses can process it when they have a valid lawful basis and satisfy other data-protection duties.

THEORY

Explain the impact on different groups

A strong response considers who benefits, who may be disadvantaged, and what follows.

AI IN RECRUITMENT

Benefit: may process many applications consistently. Risk: biased training data may discriminate unfairly. Judgement: human review and checks can reduce some risks.

SMARTPHONES & LOCATION

Benefit: maps and emergency services can use location to help. Risk: excessive location tracking exposes sensitive habits and movements.

DATA CENTRES

Benefit: online services and cloud storage support school and work. Risk: electricity, water and equipment manufacture carry environmental costs.

SOCIAL MEDIA

Benefit: families and communities can stay connected. Risk: oversharing, harmful content or intense usage can affect privacy and wellbeing.

RIGHT TO REPAIR

Benefit: longer-lived products can save money and reduce e-waste. Trade-off: repairs may be difficult or costly for some devices.

THEORY

Three laws you must recognise

The OCR specification names these laws. Explain the purpose, what actions they allow or restrict, and the impact on technology users.

DATA PROTECTION ACT 2018

Alongside UK GDPR, sets rules for processing personal data. Organisations need a valid lawful basis, appropriate security and respect for people’s rights. Consent is not automatically required if another lawful basis applies.

COMPUTER MISUSE ACT 1990

Makes specified unauthorised access to computer material and other harmful unauthorised computer acts offences. Authorised access does not give unlimited permission to alter or disrupt data.

COPYRIGHT, DESIGNS AND PATENTS ACT 1988

Protects rights in creative works, including software. Authors and rights holders can control copying and distribution, grant licences and challenge infringement.

WHAT OCR DOES NOT REQUIRE

OCR focuses on the purpose and the effects of these acts in computing scenarios—not memorising every section number, court procedure, or complete UK GDPR lawful-basis list. Licensing will receive its own full revised reference section.

THEORY · LEGAL COMPARISON

Computing legislation

Use this legislation comparison and the TIER example to link legal requirements to real effects.

Legislation	Purpose	Allows	Prohibits
Data Protection Act 2018	Protect personal data	Lawful data processing	Misuse or insecure storage of data
Computer Misuse Act 1990	Protect systems from cybercrime	Authorised access	Hacking, malware, DoS attacks
Copyright Designs and Patents Act 1988	Protect creative work	Licensed use of content	Piracy and plagiarism

IMPORTANT DATA-PROTECTION CLARIFICATION

Consent is one possible lawful basis for processing personal data, but not the only one. Organisations still need to handle data lawfully, fairly, transparently and securely.

Exam Question	T – Technology	I – Issue (Law)	E – Effect (Allows / Prohibits)	R – Result (Why it matters)
Discuss the impact of the Data Protection Act 2018 on the use of online databases by organisations.	Online databases store personal data such as names, addresses, and dates of birth.	The DPA2018 applies because this is personal data.	Allows lawful data processing and secure storage but prohibits sharing data without consent or insecure storage.	Organisations must protect data properly, reducing privacy breaches and misuse.

TIER + THE LAW

Use TIER to connect a law to what it allows or restricts, how people are affected and a justified conclusion.

KEY TERMINOLOGY

Core exam vocabulary · 1 of 2

The translucent yellow strokes are the same SVG paths and colours used in the classroom lesson pages.

Ethical issue

A question of right, wrong, fairness or responsibility when developing or using a technology.

Legal issue

Whether use of a technology complies with relevant laws and obligations.

Cultural issue

An effect on how people communicate, live, work, behave or share beliefs.

Environmental impact

An effect on energy use, resources, emissions, waste or natural surroundings.

Privacy

Control over access to personal information and how it is collected, used or shared.

Personal data

Information relating to a living person who can be identified directly or indirectly.

KEY TERMINOLOGY

Core exam vocabulary · 2 of 2

The translucent yellow strokes are the same SVG paths and colours used in the classroom lesson pages.

Data Protection Act 2018

UK legislation governing personal-data handling alongside the UK GDPR, protecting rights and setting rules.

Lawful basis

A valid legal reason for processing personal data; consent is only one possible lawful basis.

Computer Misuse Act 1990

UK law creating offences for specified unauthorised computer access and harmful unauthorised acts.

Copyright, Designs and Patents Act 1988

UK law protecting authors’ and creators’ rights over works, including computer software.

Stakeholder

A person or group affected by a decision, such as users, businesses, communities or the environment.

Balanced discussion

An extended answer explaining different effects, relevant evidence and a reasoned judgement.

WORKED EXAMPLES

Example 1 · Social-media discussion

Apply TIER to a six-mark issue question, then convert the notes into connected paragraphs.

QUESTION

Discuss the ethical, cultural and privacy issues caused by social media. (6 marks)

T — TECHNOLOGY

People post messages, images and opinions; content reaches friends and wider audiences.

I — ISSUES

Apps may collect personal data (privacy); posts may influence behaviour, expectations and relationships (cultural/ethical).

E — EFFECTS

Connecting people is beneficial, but oversharing may expose private details; online pressure may affect relationships and wellbeing.

R — RESULT

Privacy controls and responsible behaviour help, but users and platform operators share responsibility. Weigh the benefits against risks.

EXAM-WORTHY SENTENCE

“Social media can help distant communities communicate, but data collection and public posts can reduce users’ control over personal information; balanced rules and sensible privacy choices help manage those risks.”

WORKED EXAMPLES

Example 2 · A school online database

Use TIER when a question connects a named law to a realistic technology.

QUESTION

Discuss the impact of the Data Protection Act 2018 on an organisation using an online database. (6 marks)

T — TECHNOLOGY

A school database stores pupils' names and contact information to run school services.

I — LEGAL ISSUE

The school must use personal data lawfully and fairly and apply appropriate protection; the DPA 2018 operates alongside UK GDPR.

E — EFFECT

The school may need safeguards, staff training and controls on access. Weak security can harm pupils and lead to investigation.

R — RESULT

The rules help protect pupils' rights and trust, while allowing necessary data use when there is a valid lawful basis.

EXAM WARNING

Never say every lawful use of personal data needs consent: other lawful bases exist.

WORKED EXAMPLES

Example 3 · A reasoned judgement

Use this phone-battery dilemma to evaluate different stakeholders.

EXTENDED-RESPONSE SCENARIO

A phone company makes replacement batteries difficult to obtain but offers cheaper new phones. Discuss benefits and drawbacks for the company, customers and the environment. (8 marks, additional source exercise)

T — TECHNOLOGY

New phone models are offered cheaply, while replacement batteries are harder to obtain.

I — ISSUES

The company may earn more from upgrades; customers lose some repair choices; manufacturing raises environmental concerns.

E — EFFECTS

Customers may benefit from cheaper devices but pay more over time. Extra production uses energy/materials and increases e-waste.

R — JUDGEMENT

A repairable, longer-lasting phone may be preferable environmentally, although affordability and technical limitations matter. Explain why.

EXAM PRACTICE

Practice questions · 1 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q1 · 1 MARK

What is meant by an ethical issue in computing?

EXAM Q2 · 1 MARK

What is meant by privacy?

EXAM Q3 · 1 MARK

Name one law related to computer use.

EXAM Q4 · 2 MARKS

State two social or cultural issues caused by digital technology.

EXAM PRACTICE

Practice questions · 2 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q5 · 2 MARKS

State two environmental issues caused by computer systems.

EXAM Q6 · 3 MARKS

Explain one privacy issue caused by smartphones.

EXAM Q7 · 3 MARKS

Explain one ethical issue caused by artificial intelligence.

EXAM Q8 · 3 MARKS

Explain one cultural issue caused by social media.

EXAM PRACTICE

Practice questions · 3 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q9 · 4 MARKS

Describe the purpose of the Data Protection Act 2018.

EXAM Q10 · 4 MARKS

Describe what the Computer Misuse Act 1990 protects against.

EXAM Q11 · 4 MARKS

Describe what the Copyright, Designs and Patents Act 1988 protects.

EXAM PRACTICE

Practice questions · 4 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q12 · 6 MARKS

Discuss the ethical, cultural and privacy issues caused by the use of social media.

EXAM Q13 · 6 MARKS

Discuss the impact of the Data Protection Act 2018 on organisations using online databases.

EXTENSION QUESTION · 8 MARKS

A phone company makes replacement batteries difficult to obtain but offers cheaper new phones. Discuss benefits and drawbacks for the company, customers and the environment.

ANSWERS & MARK SCHEME

Indicative marking · 1 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q1 · 1 MARK

What is meant by an ethical issue in computing?

An ethical issue concerns whether using a digital technology is right, fair or responsible, including effects on people. (1)

EXAM Q2 · 1 MARK

What is meant by privacy?

Privacy is a person’s ability or right to control access to and use of information about them; it includes what organisations collect, store and share. (1)

EXAM Q3 · 1 MARK

Name one law related to computer use.

One valid law: Data Protection Act 2018, Computer Misuse Act 1990, or Copyright, Designs and Patents Act 1988. (1)

EXAM Q4 · 2 MARKS

State two social or cultural issues caused by digital technology.

Any two developed cultural/social issues, such as: online messaging can replace some face-to-face contact (1); social media can change attitudes and expected behaviour/communication (1). Accept other distinct effects. (2)

ANSWERS & MARK SCHEME

Indicative marking · 2 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q5 · 2 MARKS

State two environmental issues caused by computer systems.

Any two distinct environmental impacts, e.g.: data centres consume significant electricity (1); manufacturing/disposal creates electronic waste and uses scarce materials (1). Other supported points may include heat, water consumption or transport emissions. (2)

EXAM Q6 · 3 MARKS

Explain one privacy issue caused by smartphones.

Phones/apps may collect or infer a user’s location (1); that information can be stored or shared and used to track activities (1), so the person may lose control over who knows where they have been (1). (3)

EXAM Q7 · 3 MARKS

Explain one ethical issue caused by artificial intelligence.

An AI system trained on biased or unrepresentative data may make unfair decisions (1), such as unequal screening of job candidates (1); this could discriminate against individuals or groups (1). (3)

EXAM Q8 · 3 MARKS

Explain one cultural issue caused by social media.

Social media can shift communication away from face-to-face interactions (1), promote trends and influence language or social norms (1), and affect how people behave or maintain relationships (1). (3)

ANSWERS & MARK SCHEME

Indicative marking · 3 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q9 · 4 MARKS

Describe the purpose of the Data Protection Act 2018.

The Data Protection Act 2018 sits alongside UK data-protection law and regulates processing of personal data (1). Organisations must process information lawfully and fairly (1), protect it with appropriate security (1), and respect relevant individual rights such as access or correction (1). (4)

EXAM Q10 · 4 MARKS

Describe what the Computer Misuse Act 1990 protects against.

Computer Misuse Act 1990 makes specified unauthorised access to computer material an offence (1), including access with intent to commit further offences (1); it also covers unauthorised actions intended to impair computer operation/data (1). This protects users and organisations from attacks such as hacking (1). (4)

EXAM Q11 · 4 MARKS

Describe what the Copyright, Designs and Patents Act 1988 protects.

The Copyright, Designs and Patents Act 1988 protects original creative works including software (1); it gives rights holders control over copying/distribution (1), allows legal licensing and permitted use (1), and can make unauthorised reproduction or sharing an infringement (1). (4)

ANSWERS & MARK SCHEME

Indicative marking · 4 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q12 · 6 MARKS

Discuss the ethical, cultural and privacy issues caused by the use of social media.

Use TIER and cover several issues: T: social media allows rapid communication and sharing. I: platforms may collect personal data, while online norms, trends or overuse change how people communicate. E: positives include maintaining distant relationships, but risks include loss of privacy and reduced face-to-face interaction or pressure on wellbeing. R: explain safeguards such as controls on personal-data sharing and responsible use, with a reasoned conclusion. For six marks, explain linked impacts, not simply name categories. (6)

EXAM Q13 · 6 MARKS

Discuss the impact of the Data Protection Act 2018 on organisations using online databases.

T: organisations use online databases to store customers' personal information. I: the Data Protection Act 2018 and UK data-protection rules require lawful, fair handling of that data. E: organisations need appropriate security, transparency and a valid lawful basis (consent is not automatically required), and must respect data rights. Unlawful processing or weak security may lead to investigations, fines and loss of trust. R: compliance helps protect individuals and enables responsible use of databases. Explain both organisational obligations and consequences. (6)

EXTENSION QUESTION · 8 MARKS

A phone company makes replacement batteries difficult to obtain but offers cheaper new phones. Discuss benefits and drawbacks for the company, customers and the environment.

A developed, balanced discussion should use TIER. Technology: cheaper new smartphones, with replacement batteries difficult to obtain. Issues: customer cost/convenience, business sales/profit, product longevity, resources and waste. Effects: customers may benefit from cheaper newer models but pay more often or lose repair choice; the company may increase sales but suffer reputational damage; new manufacture uses materials/energy and creates electronic waste. Result: discuss repair access, right-to-repair approaches or trade-in/refurbishment and end with a justified judgement balancing stakeholders. This is a source extended task, not one of the original 13 table questions.

THEORY · VISUAL ANALOGY

Software licences

The recipe analogy helps explain permissions to use, change and share software.

THE RECIPE ANALOGY



SHARED RECIPE

Open source: you may read and adapt the recipe if you follow the sharing rules.



SEALED RECIPE

Proprietary: you can use the finished meal, but cannot assume permission to change the secret recipe.

Exam connection: The “recipe” represents the source code. A licence defines the permissions and restrictions. Unlike a recipe analogy, the legal terms matter: some open-source licences have conditions for sharing changes, while proprietary licences may allow use without sharing source code.

Definition: A software licence sets out the permissions and conditions for using, modifying or sharing software.

THEORY · COMPARE

Open source versus proprietary

Compare source-code permissions, not just prices or brand names.

Feature	Open source	Proprietary
Source code	Available to examine	Usually not provided
Modification	Permitted by its licence	Usually not permitted
Cost	May be free or paid	May be free or paid
Redistribution	Allowed with conditions	Often restricted
Support	Community or paid services	Vendor or external services
Examples	School coding tool / library	Commercial closed-source app

Important: Open source does not mean “always free”. Proprietary does not mean “always paid”, “always more reliable” or “always professionally supported”. Those details depend on the product and licence.

EXAM TECHNIQUE

For a recommendation, connect the licence to the user’s needs: access to source code → permitted changes → benefit to that user. Then give a realistic drawback or trade-off.

THEORY · RULES AND RIGHTS

What a software licence controls

Compare what each type of software licence permits.

Use

Who may install or run the software, and on how many devices or accounts.

Copy and share

Whether the program can be redistributed to somebody else.

Modify

Whether the user can inspect and change the source code.

Attribution and conditions

Whether credit, a copy of the licence or other conditions must accompany redistribution.

Support and updates

May be included or purchased separately; not guaranteed by licence type alone.

Remember: ?To prevent illegal copying? is only one possible purpose. A licence also tells users what they are allowed to do. Copyright still applies to both open-source and proprietary software.

THEORY · OCR COVERAGE

What the examination needs

Software licences are included in OCR J277 specification 1.6.1.

- Explain why software needs a licence and what a licence is for.
- Identify the two types: open source and proprietary.
- Compare access to source code and permission to change it.
- Recommend the suitable type of licence for a given scenario, including a benefit and a drawback.

LINK TO TIER

Software licensing can raise legal and ethical issues. Use TIER to structure an extended discussion:

Technology: software used by the school or business
Issues: licence restrictions, cost, maintenance, access
Effects: who can modify, share, support or afford it
Result: balanced judgement linked to the scenario

Study the two licence types, complete the worked recommendation, then attempt the 14 practice questions before checking the answers.

KEY TERMINOLOGY

Key terminology · 1 of 2

Learn the definitions; the yellow stroke is the same as in lessons.

Software licence

Legal terms and conditions specifying how software may be used or shared.

Source code

Program instructions in a readable form that a developer can edit.

Open source

Software whose source code is accessible and may be modified under its licence.

Proprietary

Software usually distributed without public access or permission to modify its source.

Copyright

Legal protection for an original work, including computer software.

Modification

Making changes to the original program’s source code.

KEY TERMINOLOGY

Key terminology · 2 of 2

Learn the definitions; the yellow stroke is the same as in lessons.

Redistribution

Giving copies of software or its modified versions to others.

Off-the-shelf

Ready-made software designed for a broad range of users.

Customisation

Altering a program to meet particular user or organisation needs.

Licence conditions

Rules that must be followed when installing, sharing or changing software.

Technical support

Help with installing, maintaining or troubleshooting software.

Vendor lock-in

Dependence on a supplier that makes it difficult or expensive to switch.

WORKED EXAMPLES · MODEL 1

A school wants editable code

Plan your recommendation before reading the model answer.

Six-mark scenario: A school wants students to learn how programs work and change the code. Recommend a type of software licence and justify your answer.

1 Identify the requirement

Students need to view and modify source code.

2 Choose a matching licence

Open source, because the licence allows source access and changes.

3 Explain the benefit

Pupils can investigate real algorithms, experiment and learn through editing.

4 Weigh a limitation

The school must check compatibility, support availability and licence conditions.

5 Reach a judgement

Open source is most appropriate where learning through source-code editing is the priority.

EXAM ANSWER

Choose open source. Students can inspect and modify the program code, helping them understand how it works. The school may also save on licence fees, although this is not guaranteed. It should check support and compatibility. Overall it best meets the requirement for editable code.

WORKED EXAMPLES · MODEL 2

A business needs supported software

An exam recommendation must justify a real scenario, not repeat stereotypes.

Six-mark scenario: A business needs reliable word-processing software with technical support. Recommend a type of software licence and justify your choice.

Requirement

Reliable word processing, predictable updates and accessible technical help.

Likely choice

Proprietary may be appropriate if the selected package includes a support contract.

Benefit

A named provider may offer a service agreement and compatible document formats.

Trade-off

Licence or subscription costs and reliance on the supplier can increase costs.

Important caution

Open-source applications can also have paid support, so test the actual product.

BALANCED CONCLUSION

A proprietary package may be a good choice where the vendor offers verified technical support, updates and format compatibility. The business should weigh those benefits against recurring costs and dependence on a supplier. The best choice depends on the actual products available.

EXAM PRACTICE · Practice questions Practice · Questions 1?5

Attempt every question before turning to the Answers & Mark Scheme section.

Q1 · Define · 1 mark

What is a software licence?

Q2 · Define · 1 mark

State one purpose of a software licence.

Q3 · Define · 1 mark

Name one type of software licence.

Q4 · Explain · 2 marks

State two purposes of a software licence.

Q5 · Explain · 2 marks

State two features of open source software.

EXAM PRACTICE · Practice questions Practice · Questions 6?9

Attempt every question before turning to the Answers & Mark Scheme section.

Q6 · Explain · 2 marks

State two features of proprietary software.

Q7 · Explain · 3 marks

Explain why open source software allows users to modify the program.

Q8 · Explain · 3 marks

Explain why proprietary software is often sold as off-the-shelf software.

Q9 · Explain · 3 marks

Explain one benefit of using open source software.

EXAM PRACTICE · Practice questions Practice · Questions 10?12

Attempt every question before turning to the Answers & Mark Scheme section.

Q10 · Describe · 4 marks

Describe the difference between open source and proprietary software.

Q11 · Describe · 4 marks

Describe two drawbacks of proprietary software.

Q12 · Recommend · 6 marks

A school wants students to learn how programs work and change the code. Recommend a type of software licence and justify your answer.

EXAM PRACTICE · Practice questions Practice · Questions 13?14

Attempt every question before turning to the Answers & Mark Scheme section.

Q13 · Recommend · 6 marks

A business needs reliable word-processing software with technical support. Recommend a type of software licence and justify your choice.

Q14 · Discuss · 6 marks

Discuss the benefits and drawbacks of using open source software.

ANSWERS & MARK SCHEME

Answers · Questions 1–4

Use these indicative marking points to improve your answer.

Q1 · 1 mark

What is a software licence?

A licence sets out permitted use, installation, copying and distribution of software.

Q2 · 1 mark

State one purpose of a software licence.

To grant permission for use under stated conditions; it can protect the copyright holder’s rights.

Q3 · 1 mark

Name one type of software licence.

Open source or proprietary.

Q4 · 2 marks

State two purposes of a software licence.

Specify conditions of permitted use and protect the creator’s rights or restrict unauthorised copying/distribution.

ANSWERS & MARK SCHEME

Answers · Questions 5–8

Use these indicative marking points to improve your answer.

Q5 · 2 marks

State two features of open source software.

Users can access its source code; the licence permits modification (and usually redistribution under conditions).

Q6 · 2 marks

State two features of proprietary software.

The source code is normally unavailable to users; the licence restricts modification and redistribution.

Q7 · 3 marks

Explain why open source software allows users to modify the program.

Its licence gives access to source code and permission to change it, enabling users to customise or fix features.

Q8 · 3 marks

Explain why proprietary software is often sold as off-the-shelf software.

It may be developed for a broad market, supplied as a ready-to-use package and sold or licensed without source-code modification rights.

ANSWERS & MARK SCHEME

Answers · Questions 9–11

Use these indicative marking points to improve your answer.

Q9 · 3 marks

Explain one benefit of using open source software.

It can be adapted to specific needs because developers can inspect and modify source code; licensing charges may also be lower.

Q10 · 4 marks

Describe the difference between open source and proprietary software.

Open source grants access and modification rights; proprietary usually keeps source code closed and places tighter limits on copying and modification.

Q11 · 4 marks

Describe two drawbacks of proprietary software.

Licence fees or recurring subscription costs may apply; lack of source access restricts how users can customise software.

ANSWERS & MARK SCHEME

Answers · Questions 12–14

Use these indicative marking points to improve your answer.

Q12 · 6 marks

A school wants students to learn how programs work and change the code. Recommend a type of software licence and justify your answer.

Open source: students can view and modify source code, learn by experimentation and potentially share improvements under the licence. Some packages may need extra training or support; evaluate suitability before concluding.

Marking guidance: For full credit: a justified choice, relevant benefits, an explained limitation, and a conclusion tied to the scenario.

Q13 · 6 marks

A business needs reliable word-processing software with technical support. Recommend a type of software licence and justify your choice.

Proprietary is defensible if a suitable product includes a support agreement, updates and compatibility; compare total cost and vendor dependence. Open-source software can also be supported professionally, so reliability and support must be evidenced for the selected product.

Marking guidance: For full credit: a justified choice, relevant benefits, an explained limitation, and a conclusion tied to the scenario.

Q14 · 6 marks

Discuss the benefits and drawbacks of using open source software.

Benefits: source-code access, adaptability, community review and potential licensing savings. Drawbacks: availability of specialist support, maintenance effort, compatibility and training can vary. A balanced answer reaches a context-dependent judgement.

Marking guidance: For full credit: balance several explained benefits and drawbacks and make a clear contextual judgement.